LESSON 8.10

SURFACE AREA OF RECTANGULAR PRISMS

LESSON INTRODUCTION

MA.6.GR.2.4 Given a mathematical or real-world context, find the surface area of right rectangular prisms and right rectangular pyramids using the figure's net.

LEARNING TARGETS

- I can find the surface area of a rectangular prism using a net or the figure.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.GR.4.6 Solve mathematical and real-world problems involving the surface area of three-dimensional figures limited to cylinders, pyramids, prisms, cones and spheres.

ESSENTIAL QUESTIONS

- What does the net of a rectangular prism look like?
- What is surface area?
- What strategies can you use to find surface area of a rectangular prism?
- How can a net be used to find the surface area of a rectangular prism?

LESSON NARRATIVE

In this lesson, students are introduced to finding the surface area of nets. Through guided instruction, students find the surface area of a rectangular prism using a net by first labeling given dimensions and finding the area of each face. By the end of the lesson, students show their learning by identifying all dimensions, labeling nets accurately, and finding the surface area of real-world examples of rectangular prisms.

COMPONENT SUMMARY

8.10.1 WARM-UP *(~5 MIN.)*

Which One Doesn't Belong? activity with patterns

8.10.2 GUIDED INSTRUCTION *(15-20 MIN.)*

Guided instruction on finding the surface area of rectangular prisms using nets

8.10.3 GUIDED PRACTICE *(10 MIN.)*

Find the surface area of rectangular prisms and compare student work

8.10.4 YOUR TURN! *(5 MIN.)*

Individual practice determining the surface area of rectangular prisms within real-world contexts

8.10.5 WRAP-UP *(~5 MIN.)*

Assessment on using a net to label dimensions and finding the surface area of a rectangular prism

RESOURCES

GLOSSARY

Key Terms:

- Surface area

Additional Terms:

- Net
- Face

MATERIALS

- (Optional) Four-function calculator
- (Optional) Graph paper to draw and cut out nets or preprinted nets

ADDITIONAL PREPARATION

N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may miscalculate the missing dimensions.
- Students may struggle with correctly labeling the dimensions on the nets.
- Students may forget to find the sum of all the areas of the faces or miss one of the faces.

THINGS TO CONSIDER

- Finding the surface area of three-dimensional figures is a lengthy process for students to execute. Provide different methods to finding the surface area and checkpoints throughout problem-solving to assist students with resilience. Monitor student problem-solving closely for misunderstanding.
- Reinforce, as nets and as solids, that all rectangular prisms have exactly six faces and their combined surface area should include the sum of six faces.
- Determine which problems, if any, you think your students would benefit from using a calculator on. When determining which problems, consider the difficulty of the problem and if that difficulty would be a barrier to student learning.

LITERACY AND LANGUAGE ROUTINES

Collect and Display

Consider implementing Collect and Display routines after completing the 8.10.3 Guided Practice. While the students are completing question 2, collect comments they make during their discussions about the strategies. Display these comments and others to brainstorm what these comments could be referring to when their peers made them.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

This standard emphasizes experiencing math through real-world examples. In this lesson, students explore various authentic examples of rectangular prisms and work with nets to find and label dimensions and find the total surface area in real-world contexts.

DIFFERENTIATE INSTRUCTION

Consider providing graph paper, rulers, markers, scissors, and tape for students to recreate the nets in 8.10.3 Guided Practice and 8.10.4 Your Turn!. By providing these hands-on materials and visuals, students can create prisms to scale and physically label each face as they measure and find the area, additionally helping students link the characteristics of two-dimensional nets to the characteristics of three-dimensional rectangular prisms.

8.10.1 WARM-UP: WHICH ONE DOESN'T BELONG?

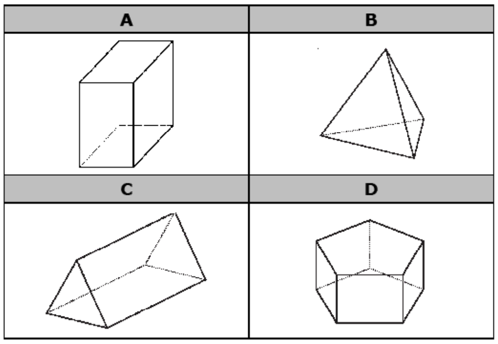
Component Overview: *Students are presented with four patterns. Students must determine which pattern does not belong with the other three and must justify their selection.*

DIFFERENTIATE INSTRUCTION

Consider a kinesthetic approach to eliciting student responses to this activity. Post a picture of each pattern, one per corner, and ask the students to stand in the corner with the picture that does not belong. Ask students to share their justifications first with their group, then to other groups in other corners. Allow students to walk between corners as an added kinesthetic approach to clarify and justify thinking.

8.10.1 Warm-Up: Which One Doesn't Belong?

1. Four figures are shown.



a. Which one doesn't belong?
Answers will vary.

b. Justify your choice using mathematics.

A is the only one where all of the faces are rectangles; A is the only rectangular prism.

B is the only one where all the faces are triangles; or the only one that doesn't include rectangular faces; or the only one that isn't a prism; or the only one that is a pyramid.

C is the only one that include rectangular and triangular faces.

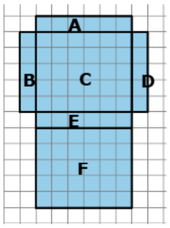
D is the only one with faces that aren't rectangles or triangles; is the only one with pentagonal faces.

8.10.2 Guided Instruction: Surface Area of a Rectangular Prism

To find the surface area of a three-dimensional object, use the nets of a shape to find the area of each face.

Surface area is the total area of the two-dimensional surfaces that make up a three-dimensional object

1. A net of a rectangular prism is shown below with each face labeled.
   a. Complete the table to find the area of each face. Each unit of the grid is 1 centimeter (cm).



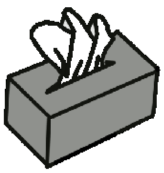
Face	Dimensions	Area
A	$1\ cm \times 6\ cm$	$6\ cm^2$
B	$1\ cm \times 5\ cm$	$5\ cm^2$
C	$5\ cm \times 6\ cm$	$30\ cm^2$
D	$1\ cm \times 5\ cm$	$5\ cm^2$
E	$1\ cm \times 6\ cm$	$6\ cm^2$
F	$5\ cm \times 6\ cm$	$30\ cm^2$

   b. How can the net and the area of the six faces be used to find the surface area of the prism?
   You can add up the areas of all the sides to find the surface area.

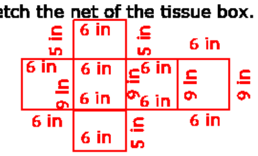
   c. What is the surface area of this rectangular prism?
   $82\ cm^2$

When calculating surface area, find the sum of the areas of each face of the figure. Use the figure's net to label the dimensions needed to calculate the area of each face.

2. A tissue box is 9 inches (in.) long, 5 in. wide, and 6 in. tall.



   a. Sketch the net of the tissue box.



   b. Label the dimensions of each face on the net.

   c. Find the area of each face. Include units.
   $45\ in^2$
   $45\ in^2$
   $54\ in^2$
   $54\ in^2$
   $30\ in^2$
   $30\ in^2$

   d. Add the area of the faces to find the surface area of the tissue box. Include units.
   $258\ in^2$

8.10.2 GUIDED INSTRUCTION: SURFACE AREA OF A RECTANGULAR PRISM

Component Overview: *Students are introduced to finding the surface area of two rectangular prisms using nets. Students are given a net on a grid and must find all the dimensions, the area of each face, and the surface area. Students are also given a real-world example of a rectangular prism and must draw the net, label it, and find the surface area. Through guided instruction and discussion, position students to identify the shape of each face and to find the area of each shape.*

ENRICHMENT

For question 1, elicit student thinking on key elements about the figure before completing the table by asking:

- How can we find the dimensions of each shape?
- How can we verify that the net of the prism is made up of rectangles and not squares?
- Since the net is on a grid, what is another method we could use to find the surface area instead of adding all the areas?

DIFFERENTIATE INSTRUCTION

Before reading the blue box explaining how to calculate surface area aloud, have students read it individually to themselves, circling any words they do not recognize or are unfamiliar with. Then define any needed words and read the description aloud. This auditory approach may be especially helpful for ELL students by frontloading key vocabulary content that is needed to access procedural knowledge on finding surface area using a net.

TEACHER MOVES

Collect and Display

After students complete question 2, prompt students to consider a different arrangement of the faces of the tissue box. Consider displaying labeled nets that are not identical to each other but still use the correct dimensions, then discuss how they are all correct. Students may point out that all the nets have six shapes. Students may also share that when folded, the nets form a rectangular prism. Point out that the arrangement of the six shapes could be different, but when folded, they still create the same rectangular prism.

8.10.3 GUIDED PRACTICE: SURFACE AREA OF A RECTANGULAR PRISM

Component Overview: *Students work in partners to solve real-world problems related to the surface area of rectangular prisms, including labeling a net's dimensions and finding the surface area by first finding the area of each face. Consider regrouping and sharing ideas at the end of 8.10.3 Guided Practice section as students compare two different strategies used to find the surface area.*

ENRICHMENT

Consider asking students these questions before completing question 1 to assist them with recognizing key elements of surface area within the real-life example:

- What is the shape of the box that Gretchen is preparing to paint?
- Why does Gretchen need to find the surface area of the box to paint it instead of the volume?

TEACHER MOVES

Deepen understanding of nets and encourage flexible thinking about representations of three-dimensional shapes by asking students to consider if Faces E and F can ONLY be in the positions given in question 1b. Position students to recognize that the two, Faces E and F, are the bases of the prism and can be attached to any of the lateral faces (A, B, C, or D) when drawing a net.

Consider providing physical examples by cutting out nets with the base Faces E and F connected to different lateral faces, then folding each net to show all four shape options create the same prism.

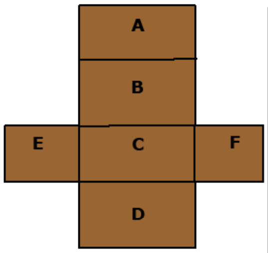
8.10.3 Guided Practice: Surface Area of a Rectangular Prism

1. Gretchen is preparing to paint a jewelry box but does not know how much paint she needs to buy.

   a. What information about the jewelry box does Gretchen need to know in order to buy paint?
   Gretchen needs to know length, width, and height.

   b. Gretchen measured the box to be 17.5 inches (in.) long, 11.5 in. wide, and 15 in. tall.

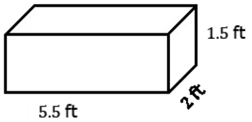
The net of the jewelry box is shown. Label the side lengths of the net. Record the dimensions in the table.



Face	Dimensions	Area
A	11.5 in. x 17.5 in.	201.25 in^2
B	15 in. x 17.5 in.	262.5 in^2
C	11.5 in. x 17.5 in.	201.25 in^2
D	15 in. x 17.5 in.	262.5 in^2
E	11.5 in. x 15 in.	172.5 in^2
F	11.5 in. x 15 in.	172.5 in^2

   c. What is the surface area of the box that Gretchen needs to buy paint for? Include units.
   1272.5 in^2

2. Jillian and Victor are finding the surface area, in square feet (ft^2), of the rectangular prism shown and each got the same answer; however, they solved the problem differently.



Jillian's Work	Victor's Work
Side 1: $5.5\text{ ft} \times 1.5\text{ ft} = 8.25\text{ ft}^2$	2 sides: $2(5.5\text{ ft} \times 1.5\text{ ft}) = 16.5\text{ ft}^2$
Side 2: $5.5\text{ ft} \times 1.5\text{ ft} = 8.25\text{ ft}^2$	2 sides: $2(5.5\text{ ft} \times 2\text{ ft}) = 22\text{ ft}^2$
Side 3: $5.5\text{ ft} \times 2\text{ ft} = 11\text{ ft}^2$	2 sides: $2(2\text{ ft} \times 1.5\text{ ft}) = 6\text{ ft}^2$
Side 4: $5.5\text{ ft} \times 2\text{ ft} = 11\text{ ft}^2$	Total Surface Area: $44.5\ ft^2$
Side 5: $2\text{ ft} \times 1.5\text{ ft} = 3\text{ ft}^2$	
Side 6: $2\text{ ft} \times 1.5\text{ ft} = 3\text{ ft}^2$	
Total Surface Area: 44.5 ft^2	

Discuss with your partner each student's strategy and how they are similar or different. Draw arrows to show connections between each student's work.

Answers will vary.

Jillian found the area of each side then add them all together. Victor knew that 2 sides that were parallel to each other would have the same measurements therefore, he multiplied the 3 different side areas by 2 and then added. They both added the side areas.

8.10.3 GUIDED PRACTICE: SURFACE AREA OF A RECTANGULAR PRISM (CONTINUED)

QUESTIONING

Extend and connect geometric concepts by prompting students to identify congruent shapes. Ask students how to determine if a shape is similar or congruent, then connect this concept to nets by identifying each pair of congruent sides and justify their reasoning. Students may respond that the top and the bottom are congruent because the dimensions are identical.

TEACHER MOVES

Collect and Display

After students have discussed each student's strategy and how they are all similar and different in question 2, consider having students write which strategy they prefer on a sticky note. Display these sticky notes on a shared classroom space and discuss which strategy is more popular. Point out that both strategies work, but the process and approach differ. Encourage students to try the strategy they did not pick on a future problem and return to this discussion after question 2 in 8.10.4 Your Turn!.

8.10.4 YOUR TURN!

Component Overview: *An image of a rectangular prism is given, and students practice finding the surface area. Students are also given a real-world scenario involving a rectangular prism and must find the surface area. While some students work individually to practice these skills, consider working with a group of students who may need additional support identifying the shape of each face, labeling the dimensions, and finding the area of each shape. Allow students to draw nets that are not identical to each other.*

DIFFERENTIATE INSTRUCTION

Consider prompting students to draw a net for question 1, then recording the area of each shape inside of the shape on their drawings of the net. By labeling the area of each face, students practice a visual strategy that will assist them with finding the surface area.

QUESTIONING

For question 1, consider challenging students by presenting the same labeled rectangular prism but turned differently, such as one prism shown horizontally and the other prism shown vertically. Ask half of the class to solve the problem using the horizontal figure and the other half of the class to solve the problem using the vertical figure. Providing the same image in two different views reinforces that surface area can be found regardless of the positioning of the rectangular prism.

TEACHER MOVES

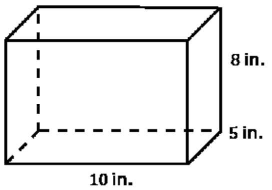
For question 2, encourage a discussion about the different methods used to find the surface area, such as with a formula, using a net to find the sum of all face areas, or finding areas of three different-sized faces and multiplying the sum by two to represent the congruent face.

Ask the students which method is easiest for them and to justify their answer. Responses could include that adding the areas from a net is the easiest because all the shapes are easiest to observe.

8.10.4 Your Turn!

1. **Darsey wants to build trinket boxes made of wood to sell at the local craft festival. A drawing of her box is shown. How much wood does Darsey need to buy for each trinket box?**

   $340\ in^2$



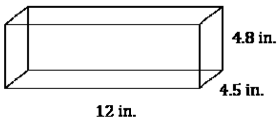
2. **A carpenter is staining planks of wood to build the top of a long picnic table. The carpenter determines the surface area of each plank in order to purchase the right amount of stain. Each plank is $\frac{3}{4}$ inches (in.) thick, $5\frac{1}{2}$ in. wide, and 72 in. long. What is the surface area of each plank?**

   $908\frac{1}{4}\ in^2$

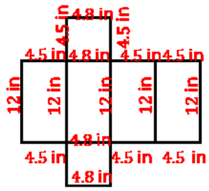
8.10.5 Wrap-Up: Ticket Out the Door

Ted is preparing a holiday display for the post office. He takes an unfolded box and uses it to determine the amount of wrapping paper needed to wrap the box.

The shipping box Ted uses for the display is shown.



1. Write the measurements on the net of the box.



2. Use the net to find the surface area of the shipping box. Include units.
   $266.4\ in^2$

8.10.5 WRAP-UP: TICKET OUT THE DOOR

Component Overview: *Students individually solve two problems by labeling a net and finding surface area within a real-world context. Consider using student responses as a formative assessment of this lesson's learning targets.*

QUESTIONING

Post these questions to reinforce the essential questions for early finishers or all students based on available time:

- How do you know what the shapes of the faces are?
- Which faces have the exact same area? How can you tell? How many faces have the same area?
- How can you use the net to find the total area of all the surfaces of this prism?